

JEE MAIN REVISION MASTER: AREA UNDER THE CURVES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 17 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Area Under The Curves — Core Principles & Governing Laws	15 PYQs	Core Principle
Standard	Area Under The Curves — Standard Calculations & Problem Archetypes	2 PYQs	Quantitative Analysis
Standard	Area Under The Curves — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Area Under The Curves — Core Principles & Governing Laws	$y=\tan x$, $y=\cos x$, $y=2x+1$, $y=x^2$
Area Under The Curves — Standard Calculations & Problem Archetypes	$y=16$, $y=\frac{3}{5}(8^{5/3}-1^{5/3})$, $y^2=x^3$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Area Under The Curves — Core Principles & Governing Laws (Very High Yield — 15 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Area Under The Curves.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Area Under The Curves.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Area Under The Curves PYQs.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "The area bounded by the curve $y=\sin((x / 3))$, x axis, the lines $x=0$ and $x=3\pi$ is..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The area of the region bounded by the curve $y=x^2$ and the line $y=16$ is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The area of the region bounded by the line $y=x$ and the curve $y=x^3$ is..." [Ans: N/A]

3.2 Area Under The Curves — Standard Calculations & Problem Archetypes (Standard Yield — 2 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Area Under The Curves.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The area of the region bounded by the curve $y^2=x^3$, the y -axis and the lines $y=1$ and $y=8$ is..." [Ans: N/A]
- **[2017 Shift] Q#4 (MCQ):** "The area of the region bounded by the curve $y=x^2$ and the line $y=16$..." [Ans: N/A]

3.3 Area Under The Curves — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Area Under The Curves.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The area of the region bounded by the curve $y^2=x^3$, the y -axis and the lines $y=1$ and $y=8$ is..." [Ans: N/A]
- **[2025 Shift] Q#2 (MCQ):** "The area bounded by the curve $y=\sin((x/3))$, x axis, the lines $x=0$ and $x=3\pi$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The area of the region bounded by the curve $y^2=x^3$, the y -axis and the lines $y=1$ and $y=8$ is

- (A) $(155/3)$ sq. units
- (B) $(93/5)$ sq. units
- (C) 93 sq. units
- (D) 155 sq. units

Correct Answer: None

Solution Analysis: The curve is $y^2=x^3$. Since the region is between the lines $y=1$ and $y=8$, it is easier to write x in terms of y . From $y^2=x^3$ we get $x=y^{2/3}$. The y -axis is $x=0$. So, for each value of y from 1 to 8, the horizontal width of the region is $x_{\text{right}} - x_{\text{left}} = y^{2/3} - 0 = y^{2/3}$. Hence, the required area is $\int_1^8 y^{2/3} dy$. Now, $\int y^{2/3} dy = \frac{y^{5/3}}{5/3} = \frac{3}{5} y^{5/3}$. So, Area $= \left[\frac{3}{5} y^{5/3} \right]_1^8 = \frac{3}{5} (8^{5/3} - 1^{5/3})$. Since $8^{1/3} = 2$ we have $8^{5/3} = 2^5 = 32$. Thus, Area $= \frac{3}{5} (32 - 1)$.

Benchmark Problem 2: KCET 2025 (Q#2)

The area bounded by the curve $y=\sin((x/3))$, x axis, the lines $x=0$ and $x=3\pi$ is

- (A) 9 sq. units
- (B) $(1/3)$ sq. units
- (C) 6 sq. units
- (D) 3 sq. units

Correct Answer: None

Solution Analysis: The required area is found by integrating the function $y=\sin((x/3))$ from $x=0$ to $x=3\pi$. This gives the definite integral: $\int_0^{3\pi} \sin((x/3)) dx$ To integrate $\sin((x/3))$, we use substitution. The integral becomes $-3\cos((x/3))$. Now substitute the limits 0 and 3π : $\int_0^{3\pi} \sin((x/3)) dx = (-3 \cdot \cos(x/3)) \Big|_0^{3\pi} = (-3(-1)) - (-3(1)) = 6$ So, the total area is 6.